MUHAMMAD AHMAD

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PROFESSIONAL SUMMARY

Computer Science undergraduate (8th semester, Lahore Garrison University, GPA 3.5) seeking an **Associate Software Developer** position. Has built and deployed full-stack applications with the **MERN stack** (React.js, Node.js, Express.js, MongoDB), including secure REST APIs with JWT access/refresh tokens, role-based access control, and cloud deployment on Vercel, Render and MongoDB Atlas. Also builds AI-powered web apps with **FastAPI**, Vision Transformers and RAG. Strong in JavaScript (ES6+), OOP, design patterns, Data Structures & Algorithms and Git, with a focus on clean, maintainable code.

TECHNICAL SKILLS

Front-End: React.js, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Responsive Web Design
Back-End: Node.js, Express.js, FastAPI, REST API Design, JWT Auth (access/refresh tokens), bcrypt, Joi/Zod Validation, Rate Limiting, Nodemailer
Databases: MongoDB, Mongoose, MongoDB Atlas, MySQL, SQL Server, Entity Framework, Database Design, SQL Queries
Cloud & Tools: Vercel, Render, Cloudinary, Git, GitHub, VS Code, Visual Studio
Languages: JavaScript, Python, C++, C#
Software Engineering: OOP, Design Patterns, Software Architecture, Data Structures & Algorithms, Debugging, Full-Stack Development
AI / ML: Vision Transformer (ViT), Retrieval-Augmented Generation (RAG), Computer Vision, OpenCV, MediaPipe

PROJECTS

AI-Based Crop-Guard Web Platform (Final Year Project) | *React.js, FastAPI, Vision Transformer, RAG*
- Building a full-stack platform for real-time crop disease identification and post-harvest storage advisory.
- Trained a **Vision Transformer** disease-detection model reaching **92% accuracy**, served through a FastAPI endpoint; users upload a crop image and get the diagnosis with a **heatmap** highlighting affected regions.
- Built a **RAG-based chatbot** that answers disease-treatment and storage questions from a curated knowledge base.
- Developed a **community forum** with an **AI toggle** that lets users request AI-generated answers on posts.

CampusEvents – University Society & Event Management Platform | *MERN, JWT, Tailwind CSS, Cloudinary, Nodemailer*
- Built and deployed a full-stack platform for university societies with **role-based access** for Admin, Organizer and Student.
- Implemented secure auth: **JWT access/refresh token rotation**, bcrypt hashing and HTTP-only cookies.
- Designed **25+ REST API endpoints** with Joi/Zod validation, centralized error handling and rate limiting.
- Added event search, category/date filters and **server-side pagination**; registration with seat limits, waitlists and automated email confirmations/reminders via Nodemailer.
- Integrated Cloudinary image uploads with type/size validation; built an admin dashboard with per-event and per-society registration stats and organizer approval workflows.
- Deployed front end on **Vercel**, API on **Render** and database on **MongoDB Atlas** with environment-based config.

Movie Lister – Full-Stack Movie Discovery App | *React.js, TMDB API, REST backend*
- Integrated the **TMDB API** to show posters, ratings and descriptions with live **search** and genre/year **filters**.
- Built a **favorites list** persisted through my own backend API, in a UI inspired by modern streaming platforms.

Sign Language Translator | *Python, OpenCV, MediaPipe*
- Built a webcam-based recognizer that detects hand landmarks and classifies ASL alphabet (A–Z) gestures into text.
- Added temporal smoothing, confidence thresholding and text-to-speech output for stable, accessible predictions.

Student Registration & Records System | *React.js, MySQL*
- Developed a dynamic multi-step registration form in React.js based on gathered requirements.
- Backed it with a MySQL CRUD application (add, update, delete, view) on a normalized student schema I designed.

EXPERIENCE

Full-Stack Development, Academic & Personal Projects
- Designed, built and deployed features end-to-end across React.js front ends, Node.js/Express and FastAPI back ends, and MongoDB/MySQL databases.
- Debugged front-end logic, API routes and database queries; used Git/GitHub for version control on solo and team projects.

EDUCATION

Bachelor of Science in Computer Science, Lahore Garrison University, Lahore | 2023–2027 | GPA: 3.5
Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, Database Systems (DBMS), Software Architecture